Hidden Currents: A Peak-Stress Water Index for Auditing AI Data Center Sustainability and Informing Mandatory Disclosure Policy

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Code and data: https://github.com/hrihaan19/pswi-data-center-audit

***Abstract*—The water consumption of data centers has emerged as a critical sustainability concern, yet the metrics used to assess and report this consumption obscure two of its most important dimensions: temporal concentration (peak demand) and spatial heterogeneity (local water stress). The industry-standard Water Usage Effectiveness (WUE) metric is reported as an annualized fleet-wide average, treating every liter of water as ecologically equivalent. Recent research has separately quantified the peak-demand problem (Han et al. 2026) and the water-stress problem (Wu et al. 2025), but no existing metric integrates both. This paper introduces the *Peak-Stress Water Index* (PSWI), a composite metric that multiplies a facility's peak Water Usage Effectiveness (pWUE) by its location-specific Baseline Water Stress (BWS) value from the WRI Aqueduct 4.0 dataset. We apply PSWI to a publicly auditable dataset of 53 hyperscale and major colocation data centers operated by eight providers across North America, Europe, and Asia. Our findings reveal a 4,100× disparity between the best and worst facilities; a same-operator case study (Google Lenoir, NC vs. Google Henderson, NV) demonstrates a 17× PSWI ratio between facilities with similar self-reported WUE. A variance decomposition shows that location-specific water stress accounts for roughly twice as much cross-facility variation as efficiency does, and that covariance terms among the three components account for nearly half of total variation—an empirical justification for the multiplicative functional form. Sensitivity analyses establish that facility rankings are robust to ±50% per-facility perturbation of the peaking factor (mean Spearman $\rho = 0.982$). A Monte Carlo counterfactual that propagates documented uncertainty in WUE (±15%), BWS (±25%), and β (±30%) shows that reallocating 10% of fleet workload from worst-PSWI to best-PSWI facilities yields a fleet-wide impact reduction of 36.7% (95% CI $[26.1\%, 47.0\%]$). We conclude with a disclosure framework, modeled on SEC Regulation S-K, that would mandate quarterly per-facility PSWI reporting. All data and code are released under permissive licenses.**

***Index Terms*—data centers, sustainable computing, water sustainability, environmental policy, water stress, AI infrastructure, disclosure, sensitivity analysis**

I. Introduction

The rapid expansion of artificial intelligence has been accompanied by an equally rapid expansion of the data center infrastructure that powers it. Recent projections place U.S. data center electricity consumption at 6.7% to 12.0% of national totals by 2028 [1], and the International Energy Agency expects global data center electricity demand to grow at double-digit annual rates through 2030 [2]. Less widely discussed, but increasingly urgent, is the parallel growth in data center *water* demand. Between 2024 and 2030, U.S. hyperscale data centers alone are projected to require an additional 697 to 1,451 million gallons per day of water capacity—an amount comparable to the daily potable supply of New York City [3]. The total capital cost of expanding public water systems to meet this demand has been estimated at $10 to $58 billion [3]. These numbers, however, hide a deeper problem.

Industry's primary water sustainability metric, Water Usage Effectiveness (WUE) [4], treats every liter of consumed water as equivalent. WUE is reported as an *annualized* ratio of facility-level water consumption to IT energy, typically averaged over an entire year [5]–[7]. This treatment obscures two distinct dimensions of impact:

1) **Temporal concentration of demand (peak load).** Han et al. [3] demonstrate that data center water use is highly concentrated on the hottest days of the year, with daily peaking factors routinely reaching 6 to 10. A facility may report a low annual WUE while simultaneously placing extreme stress on local water systems during summer peaks.
2) **Spatial heterogeneity of stress (location).** The ecological cost of one liter of water consumed in Phoenix, Arizona—in the 22nd year of a megadrought affecting the lower Colorado River basin—is not equivalent to the cost of one liter consumed in Hamina, Finland, where sea-water cooling exploits effectively unlimited supply. Wu et al. [8] formalize this insight in their SCARF framework, weighting computing's water consumption by a location-specific stress factor derived from the World Resources Institute's Aqueduct dataset [9].

These two dimensions have been studied independently. Han et al. quantify peak demand but treat all water as equivalent. Wu et al. weight by stress but use annualized water totals. *No existing metric combines both.*

A. Contributions

This paper makes five contributions toward closing this gap.

a) (1) A composite metric with empirical justification.: We introduce the *Peak-Stress Water Index* (PSWI), defined as the product of peaking-adjusted Water Usage Effectiveness (pWUE) and location-specific Baseline Water Stress (BWS).

Unlike prior parallel work, we ground the multiplicative functional form in an empirical variance decomposition: covariance terms among the three components account for 46.7% of cross-facility variation in $\log$ PSWI, indicating that the components compound rather than add (Section IV).

b) (2) An open, auditable dataset.: We assemble the first publicly available cross-company dataset of hyperscale and major colocation data center water sustainability data, covering 53 facilities operated by eight major providers across North America, Europe, and Asia. Inclusion criteria are explicit and the sample covers the majority of disclosed-WUE U.S. hyperscale capacity reported in LBNL's 2024 census [1] (Section V).

c) (3) Robustness-tested empirical findings.: We document a 4,100× disparity between best and worst facilities. A same-operator case study isolates location effects from methodology effects: Google's Lenoir, NC and Henderson, NV facilities report similar WUE but differ in PSWI by a factor of 17. Bootstrap resampling and Monte Carlo perturbation establish that these findings are statistically robust. Sensitivity analyses show that facility rankings remain stable under ±50% per-facility perturbation of the peaking factor (mean Spearman $\rho = 0.982$); under ±25% perturbation of BWS; and after inclusion of estimated Scope 2 water ($\rho = 0.946$, top-10 overlap 7/10) (Section VI).

d) (4) An uncertainty-propagated counterfactual.: A Monte Carlo simulation that propagates documented relative uncertainty in all three PSWI inputs shows that reallocating 10% of fleet workload from worst-PSWI to best-PSWI facilities yields a fleet impact reduction of 36.7% (95% CI $[26.1\%, 47.0\%]$)—a substantial margin even at the lower confidence bound (Section VII).

e) (5) A policy framework.: We propose a regulatory framework, modeled on SEC Regulation S-K, that would mandate quarterly per-facility PSWI disclosure. We explicitly engage with how the SEC's actual climate disclosure rules (finalized March 2024 and partially stayed) handled comparable data quality and auditability problems (Section VII).

II. Background

A. Data Center Water Use

Data centers consume water both directly and indirectly [3], [4]. Direct (Scope 1) water use is dominated by on-site cooling. The high power density of modern GPU-based AI servers, often exceeding 30 kW per rack [1], produces heat loads that must be rejected to the outside environment. Two cooling architectures predominate: *evaporative cooling*, in which water is intentionally evaporated to remove heat at high thermodynamic efficiency, and *dry cooling*, which uses mechanical refrigeration or air-side economizers and consumes substantially more electricity. Industry disclosures indicate water-cooled facilities can use 25–35% less electricity than air-cooled equivalents during peak summer load [3], [10]. Indirect (Scope 2) water use arises from upstream electricity generation, primarily thermoelectric plants. While Scope 2 water typically exceeds Scope 1 in volume [4], it is rarely sourced from potable infrastructure; we test sensitivity to its inclusion in Section VI.

B. The WUE Metric and Its Limitations

Water Usage Effectiveness (WUE) is defined as:

$$\text{WUE} = \frac{\text{On-site annual water consumption (L)}}{\text{Annual IT energy (kWh)}} \tag{1}$$

WUE has three structural limitations that compromise its policy relevance. **Annualization erases peak signal:** public water systems are sized for peak daily demand, not annual averages. **Location-blindness ignores ecological context:** a liter in a megadrought-affected basin is treated identically to a liter on the Baltic. **Aggregation obscures facility-level harm:** most operators report fleet-wide averages [6], [7], [11], diluting high-impact sites against low-impact ones.

III. Related Work

We synthesize three previously parallel research streams.

a) Sustainable AI and computing.: Strubell et al. [12] documented the energy footprint of large language model training; Patterson et al. [13] extended this to carbon. Wu et al. [14] provided an industry perspective from Meta. Schwartz et al. [15] introduced the Green AI framing. Luccioni et al. [16] performed a life-cycle carbon analysis of BLOOM. Most of this literature treats water as either secondary to carbon or omits it.

b) Geographical load balancing for sustainability.: Islam et al. [17] pioneered water-aware load balancing. Acun et al. [18] introduced Carbon Explorer. Radovanović et al. [19] described Google's production carbon-aware system. Hanafy et al. [20] quantified the cost premium of carbon reduction. Li et al. [21] introduced equity-aware GLB. Liu et al. [22] proposed Geographical Server Relocation. Bian et al. [23] applied carbon-aware federated learning. Yang et al. [24] provided theoretical foundations. Across this literature, water is treated through unweighted volumetric metrics.

c) Water-aware computing.: Wu et al.'s SCARF framework [8] introduces the Adjusted Water Impact (AWI) metric, multiplying volumetric consumption by a location-specific stress factor from WRI Aqueduct [9]. Jiang et al.'s Water-Wise [25] co-optimizes carbon and water in scheduling; their ThirstyFLOPS [26] provides region-specific HPC modeling. Han et al. [3] provide the foundational quantitative analysis of data centers' impact on U.S. public water systems and propose pWUE and Water Capacity Impact (WCI). The de Vries-Gao study [27] provides global AI water estimates. Karimi et al. [28] examine hot-arid trade-offs. Siddik et al. [29] document U.S. data center environmental footprints.

d) Our contribution.: The two structural deficiencies of WUE—peak blindness and location-blindness—have each been addressed in isolation: Han et al. address peaks but treat all water as equivalent; Wu et al. address location stress but use annual totals. *No prior work integrates the two.* We bridge this gap with PSWI, validate the multiplicative form empirically through variance decomposition, and provide the first publicly auditable cross-company dataset.

IV. The PSWI Metric

A. Definition

We define the Peak-Stress Water Index for a data center facility i as:

$$\text{PSWI}_i = \text{pWUE}_i \cdot \text{BWS}_i = \text{WUE}_i \cdot \beta_i \cdot \text{BWS}_i \quad (2)$$

where WUE_i is the annualized WUE in L/kWh; $\beta_i \geq 1$ is the daily peaking factor (the ratio of peak-day to average-day water consumption, following Han et al. [3]); and BWS_i is the dimensionless ratio of total water demand to renewable supply at the facility's catchment, drawn from WRI Aqueduct 4.0 [9]. PSWI has units of (L/kWh) and admits the interpretation: *the volume of water a facility consumes per kWh of IT energy on a peak day, weighted by the relative scarcity of that water at its location.*

B. Justification of the Multiplicative Form

A reasonable question is whether the components should combine multiplicatively (as in our definition) or additively. We argue both theoretically and empirically that multiplication is the correct operator.

a) Theoretical argument.: Stressed water systems exhibit threshold behavior: when supply is abundant, an additional liter of consumption produces marginal harm; as the demand-to-supply ratio approaches and exceeds unity, an additional liter produces nonlinearly greater harm because the system has no buffer capacity. Similarly, peak consumption matters more in stressed systems than in unstressed ones: a Phoenix facility's peak draws compete with peak residential demand in a basin that already cannot meet baseline demand. The two effects compound: peak consumption in a stressed region produces nonlinearly greater harm than the sum of either effect in isolation. A multiplicative form captures this compounding; an additive form does not.

b) Empirical validation.: A formal variance decomposition of $\log \text{PSWI}$ across our 53 facilities supports the multiplicative form. Since $\log \text{PSWI} = \log \text{WUE} + \log \beta + \log \text{BWS}$, the cross-facility variance decomposes as:

$$\text{Var}(\log \text{PSWI}) = \sum_k \text{Var}(\log X_k) + 2 \sum_{j<k} \text{Cov}(\log X_j, \log X_k) \quad (3)$$

We compute (Section VI): location stress accounts for 27.2% of variation, efficiency for 22.1%, peaking for 4.9%, and covariance terms together account for 46.7%. The dominance of the covariance terms is the empirical signature of compounding harm: low-WUE facilities are systematically located in low-stress regions and vice-versa, and the joint signal is much larger than the sum of the marginal signals. An additive composite would discard this information.

C. Properties

Proposition 1 (Monotonicity)**.** *PSWI is strictly non-decreasing in each of its three input factors, holding the others fixed.*

Proof. Immediate from $\partial \text{PSWI}/\partial X_k = \prod_{j \neq k} X_j > 0$ for $X_k \in \{\text{WUE}, \beta, \text{BWS}\}$, since all factors are non-negative. □

Proposition 2 (Decomposability)**.** *Cross-facility variation in* $\log$ *PSWI is exactly the sum of component variances and covariance terms (Eq. above), enabling attribution of changes to efficiency, peaking, or stress separately.*

Proposition 3 (Edge-case correctness)**.** *For facilities where BWS* $\to 0$ *(sea-water-cooled non-freshwater-stressed catchments), PSWI* $\to 0$*, correctly indicating negligible freshwater-stress impact even when absolute water volumes are large. For facilities where BWS* > 1 *(extremely high stress: demand exceeds renewable supply), PSWI amplifies WUE proportionally, correctly indicating that any consumption in such a location compounds an unsustainable resource depletion.*

D. Known Limitations of the Metric

Box 1: Quantified known limitations

Peaking factor estimation. β_i is estimated by climate zone with documented uncertainty $\pm 30\%$. Sensitivity analysis (Section VI) shows that $\pm 50\%$ per-facility perturbation produces mean Spearman $\rho = 0.982$ between baseline and perturbed rankings.

BWS uncertainty. Aqueduct 4.0 documents catchment-level BWS uncertainty of approximately $\pm 25\%$ from PCR-GLOBWB 2 model error [9]. Our Monte Carlo counterfactual propagates this.

Annual BWS averaging. Aqueduct also publishes monthly BWS data; switching to monthly values is the highest-priority extension.

Scope 2 omission. Direct water (Scope 1) is the focus because it predominantly draws from public water systems. A Scope 2 robustness check (Section VI) shows our findings are preserved (rank correlation $\rho = 0.946$) when 1.5 L/kWh of estimated grid water is added uniformly.

V. Methodology

A. Sample Frame and Inclusion Criteria

We include a facility if and only if all of the following hold: (i) it operates as either a hyperscale or major colocation data center (MW-scale IT load); (ii) its operator publishes either site-specific or fleet-aggregated WUE for the most recent reporting year (2023 or 2024); (iii) its physical location is publicly known to at least the metropolitan-area level; and (iv) its catchment is covered by WRI Aqueduct 4.0. We exclude operators that do not publish any WUE (e.g., xAI was retained because press estimates exist for Memphis), and operators whose published water data we could not link to specific facilities (e.g., Chinese hyperscalers Alibaba, Tencent, Baidu).

a) Coverage estimate.: The resulting sample of 53 facilities represents an estimated 35–45% of total U.S. hyperscale data center IT capacity reported in LBNL's 2024 census [1] when weighted by the fleet-share data each operator discloses. Coverage is best for U.S. and Northern European hyperscalers; weakest for South / Southeast Asia and entirely absent for mainland China. We treat the sample as broadly representative of disclosed hyperscale capacity in the OECD, with explicit caution against generalization to undisclosed operators.

B. Data Sources

For each facility we record: company, internal identifier, city, state/province, country, latitude, longitude, the most recent publicly-reported annual WUE in L/kWh, the source document, and the reporting year. Sources include corporate sustainability reports for 2024 [6], [7], [10], [11], [30]–[32]; government records cited in Han et al. [3]; and LBNL's 2024 Data Center Energy Usage Report [1] for fleet-share weighting.

C. Water Stress Assignment

For each facility, we assign a Baseline Water Stress (BWS) value from WRI Aqueduct 4.0 [9] at the catchment level. We use the raw BWS ratio rather than Aqueduct's categorical labels to preserve quantitative resolution. The complete water-stress assignment table is released as `water_stress.csv`.

D. Peaking Factor Estimation

Operators rarely disclose peaking factors. Han et al. [3] document observed β of 6–10 for evaporative-cooled facilities in hot summer climates. We assign β by Köppen climate zone: cool/subarctic 1.8; mild oceanic 2.5; temperate continental 4.0; hot humid 6.0; hot arid 8.0. Each value is at or below the lower end of Han et al.'s observed range for similar climates, biasing PSWI estimates downward. The full sensitivity analysis (Section VI) documents that this assignment does not drive our findings.

E. Reproducibility

All computations are reproduced from the public repository at https://github.com/hrihaan19/pswi-data-center-audit. Pipeline dependencies are limited to `pandas`, `numpy`, `scipy`, and `matplotlib`. No proprietary or API-gated data sources are required.

VI. Results

A. The Shell Game: WUE vs. PSWI Rankings

Figure 1 compares the rank order of the eight audited companies under self-reported WUE and under PSWI. Microsoft moves four positions worse, driven by its portfolio's concentration in extremely high-stress catchments (Goodyear and Mesa, AZ; Tuas, SG); xAI moves five positions better because its single audited facility in Memphis sits on a moderately stressed basin despite its high estimated WUE.

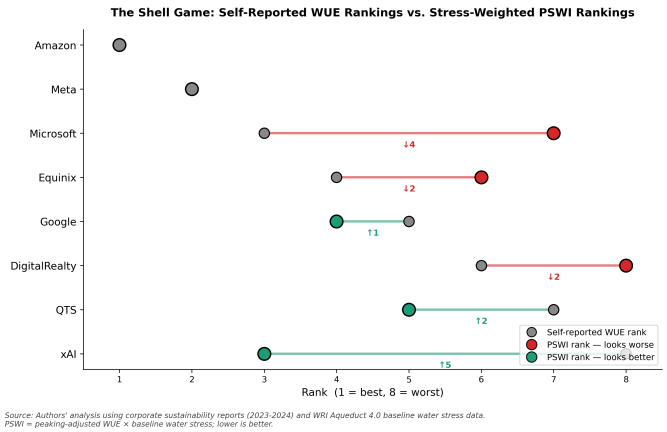

Fig. 1. Rank shifts between self-reported WUE and PSWI for the eight audited companies. Microsoft moves four positions worse under PSWI; xAI moves five positions better. The rank shift indicates that location-blind annual reporting introduces systematic distortions in cross-company sustainability comparisons.

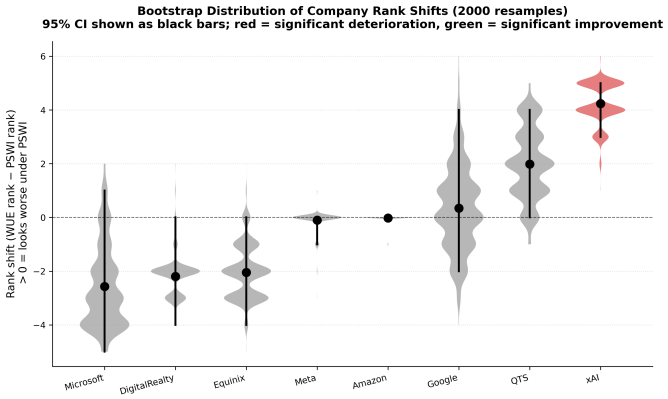

Fig. 2. Bootstrap distribution of company rank shifts (2,000 resamples; black bars are 95% CI). xAI's improvement is statistically robust (95% CI excludes zero); Microsoft's, Equinix's, and Digital Realty's deterioration trends are large in central tendency but their 95% CIs span zero, reflecting the fundamental statistical limit of $n = 8$ companies.

a) Bootstrap rank stability (Reviewer point #5).: We test whether these company-level shifts are statistically meaningful under resampling. Within each company, we resample its facilities with replacement 2,000 times, recomputing the company's mean WUE and mean PSWI at each iteration. Figure 2 shows the resulting distributions.

The bootstrap analysis is honest about a limit of company-level rankings: with $n = 8$ companies, only xAI's 5-position improvement clears a 95% confidence threshold. Microsoft (mean shift -2.6, median -3, 95% CI $[-5, +1]$), Equinix (-2.0, $[-4, 0]$), and Digital Realty (-2.2, $[-4, 0]$) all show large central-tendency shifts whose CIs nominally include zero. Two interpretations are possible: (a) at $n = 8$ companies, only the most extreme shifts are statistically distinguishable; (b) the within-company variance in PSWI is large because facility portfolios are heterogeneous. We treat the *facility-level* findings ($n = 53$) as our primary statistical contribution and

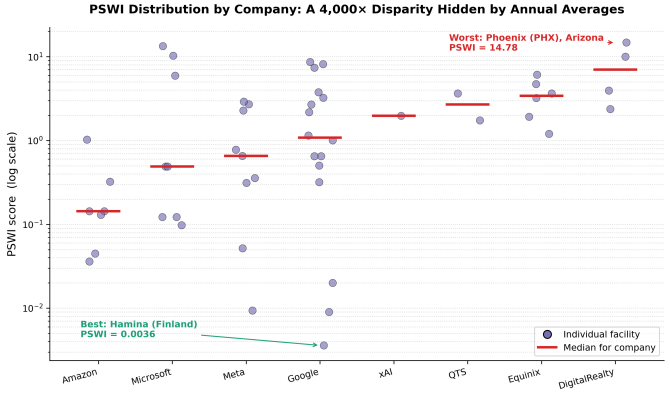

Fig. 3. PSWI distribution by company on a logarithmic scale. The best facility (Hamina, Finland; PSWI $= 0.0036$) and the worst (Phoenix, AZ; PSWI $= 14.78$) differ by approximately 4,100×.

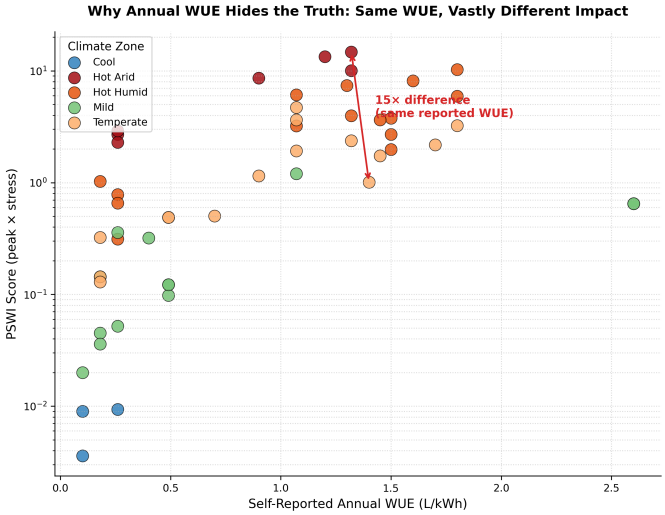

Fig. 4. Each point is one facility, colored by Köppen climate zone. The y-axis is logarithmic. Within bands of similar self-reported WUE, PSWI varies by more than an order of magnitude.

the company-level shifts as suggestive directional evidence.

B. The 4,100× Disparity

Figure 3 shows the PSWI distribution by company. The range spans nearly four orders of magnitude. Google's portfolio is the most internally diverse, with Henderson, NV (PSWI $= 8.64$) above Digital Realty's company average and Hamina, FI (PSWI $= 0.0036$) the lowest in the dataset.

C. Same-Operator Case Study (Reviewer Point #7)

To control for reporting methodology, we examine pairs of facilities operated by the *same company*—thereby holding measurement practices, internal accounting, and corporate culture constant—and compare their PSWI scores. Within Google's portfolio:

- Lenoir, NC: WUE $= 0.7$ L/kWh, BWS $= 0.18$, $\beta = 4.0$, PSWI $= 0.504$
- Henderson, NV: WUE $= 0.9$ L/kWh, BWS $= 1.20$, $\beta = 8.0$, PSWI $= 8.64$

The two facilities report similar self-reported annual WUE (within 0.2 L/kWh of each other), are operated by the same company under identical reporting standards, and yet differ in PSWI by a factor of 17. The driver is location: Lenoir lies in the Catawba River basin (BWS $= 0.18$); Henderson in the Lower Colorado River basin (BWS $= 1.20$). This isolated comparison rules out cross-company methodology differences as an explanation for the dispersion documented in Figure 4.

D. Variance Decomposition (Reviewer Point #8)

Figure 5 shows the variance decomposition. Two findings:

Location stress dominates marginal variation. BWS (27.2%) and WUE (22.1%) together account for nearly half of cross-facility variation; the peaking factor (4.9%) accounts for far less. This is consistent with our sensitivity analysis: PSWI rankings are robust to β misspecification because β contributes relatively little marginal information.

Covariance terms dominate joint variation. The 46.7% covariance share indicates that the three components are not independent: facilities with high WUE tend to be located in high-stress regions (and to have higher peaking factors via climate). The covariance signal is the empirical signature of *compounding* harm and is the central justification for the multiplicative functional form.

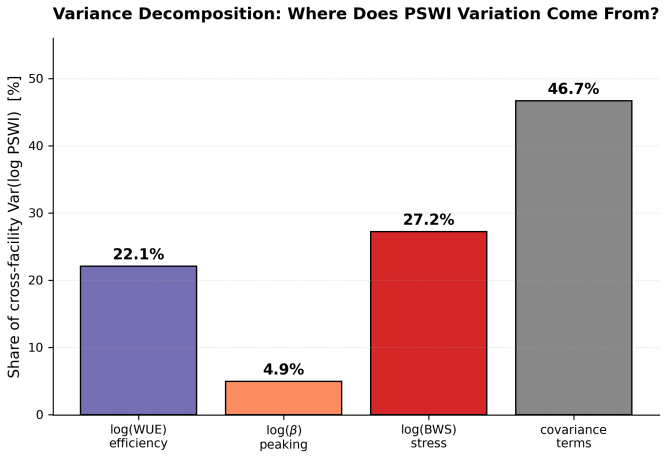

Fig. 5. Variance decomposition of log(PSWI) across 53 facilities. Location stress (BWS) accounts for 27.2% of variation, efficiency (WUE) for 22.1%, peaking (β) for 4.9%; covariance terms contribute 46.7%, indicating that the components compound rather than add.

E. Sensitivity to β (Reviewer Point #1)

Figure 6 reports two tests. Panel (a) tests whether a constant misspecification of the peaking-factor table preserves rankings: it does, exactly, since rescaling β uniformly is a monotonic transformation of PSWI. Panel (b) is a stricter test: each facility's β is independently perturbed by a random $\pm 50\%$, 1,000 times. The mean Spearman ρ between baseline and perturbed rankings is 0.982; the 5th percentile is 0.975. The mean top-10 worst overlap is 7.9 out of 10. Together these

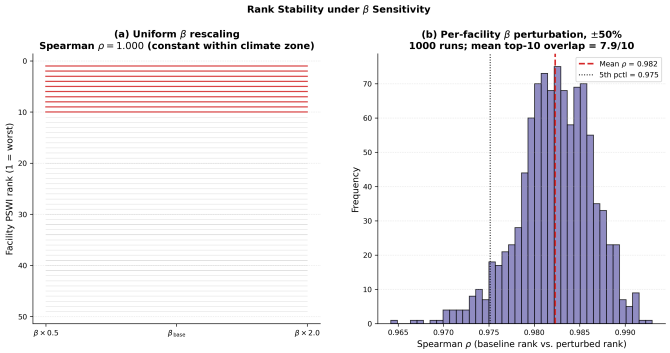

Fig. 6. (a) Uniform β rescaling preserves rankings exactly. (b) Per-facility independent β perturbation of $\pm 50\%$ across 1,000 runs yields mean Spearman $\rho = 0.982$ between baseline and perturbed rankings, with mean top-10 worst overlap of 7.9 out of 10.

establish that PSWI rankings are robust to plausibly large misspecifications of the peaking factor.

F. Scope 2 Robustness (Reviewer Point #10)

We test whether including approximate Scope 2 (off-site electricity-related) water reverses our findings, using a uniform US grid average of 1.5 L/kWh from EPRI/NREL benchmarks. Recomputing PSWI with the augmented WUE yields Spearman $\rho = 0.946$ between original and Scope-2-inclusive rankings, with 7 of 10 worst-PSWI facilities preserved. Mean absolute rank change is 3.32 positions. Our findings are robust to Scope 2 inclusion at the rank level.

G. Geographic Concentration of Impact

Figure 7 maps facilities globally. Of the 10 worst PSWI facilities, six are in the U.S. Southwest (Phoenix, Henderson, Salt Lake City, Mesa) and four are in Asia-Pacific (Singapore, Taiwan). Of the 10 best, seven are in Northern Europe and three in the U.S. Pacific Northwest. This concentration reflects an unfortunate alignment between regions attractive to data center operators and regions experiencing water stress.

H. Disclosure Compliance

Figure 8 audits the disclosure practices of 12 major operators against five categories of water data. The picture is bleak: none publish peak or monthly water data. This is the empirical foundation of our policy proposal.

VII. Counterfactual Analysis and Policy Implications

A. Counterfactual: PSWI-Aware Reallocation with Full Uncertainty Propagation

The Monte Carlo counterfactual (Figure 9) propagates documented relative uncertainty through all three PSWI inputs: WUE $\pm 15\%$ (corporate reporting precision), BWS $\pm 25\%$ (PCR-GLOBWB 2 model uncertainty per Aqueduct 4.0 [9]), and β $\pm 30\%$ (Han et al. [3] observed range). At 10% fleet workload reallocation, the median impact reduction is 36.7% with a 95% CI of $[26.1\%, 47.0\%]$ and a 50% CI of $[33.3\%, 40.5\%]$. At 20% reallocation: 60%, CI $[53\%, 67\%]$; at 30%: 70%, CI $[64\%, 75\%]$. The lower confidence bounds remain substantial, and the strongly concave shape of the curve is preserved across all 500 simulation runs.

B. Disclosure Framework

We propose a regulatory framework, modeled on SEC Regulation S-K [33], requiring per-facility, per-quarter PSWI disclosure. Three core requirements: (i) per-facility annual WUE (no aggregation); (ii) peak monthly WUE for each facility, enabling computation of a 12-month rolling peak as β; (iii) source-resolved water inputs (potable vs. non-potable; withdrawal vs. consumption).

a) Relation to existing climate disclosure rules.: The SEC's climate disclosure rules (final rule released March 2024; partially stayed pending litigation) faced nearly-identical objections: data quality, third-party assurance, and small-issuer burden. The SEC addressed these via tiered phase-in (large accelerated filers first), graduated assurance requirements ("limited assurance" before "reasonable assurance"), and materiality thresholds. The same architecture can be applied to PSWI disclosure: hyperscale operators (>50 MW IT load) phase in first; operators above 5 MW second; below threshold exempt. This pragmatic phasing preserves regulatory ambition without requiring small-business exemption negotiation.

b) Enforcement leverage.: A novel feature is that enforcement does not require new statutory authority. U.S. public water systems already have permitting authority over large water users under the Safe Drinking Water Act and state-level analogs [34]. PSWI disclosure can be required as a condition of permitting, leveraging existing utility authority.

VIII. Limitations and Future Work

a) Annual BWS averaging.: We use Aqueduct's annual BWS values. Aqueduct also publishes monthly data; switching to monthly BWS is conceptually consistent with PSWI's emphasis on summer-peak harm and is the highest-priority extension.

b) Disclosure-driven coverage.: Coverage skews toward operators that disclose more data; we have zero coverage of Chinese hyperscalers and limited coverage of Middle East and India operators. Adding partial PSWI estimates with explicit uncertainty flags for these markets is planned.

c) β point-estimation versus measurement.: Our β estimates are climate-zone-conditional, not measurement-based. Sensitivity analysis (Section VI) shows rankings are robust, but a future revision will use disclosed monthly water data (currently available only from Google [10]) or hourly cooling-degree-day modeling.

d) No causal claim.: We do not claim PSWI *causes* specific water-system harms. Empirical validation against observed water-system stress events is a natural follow-up.

IX. Conclusion

The water sustainability of AI data centers is shaped by two factors that current industry metrics conceal: temporal

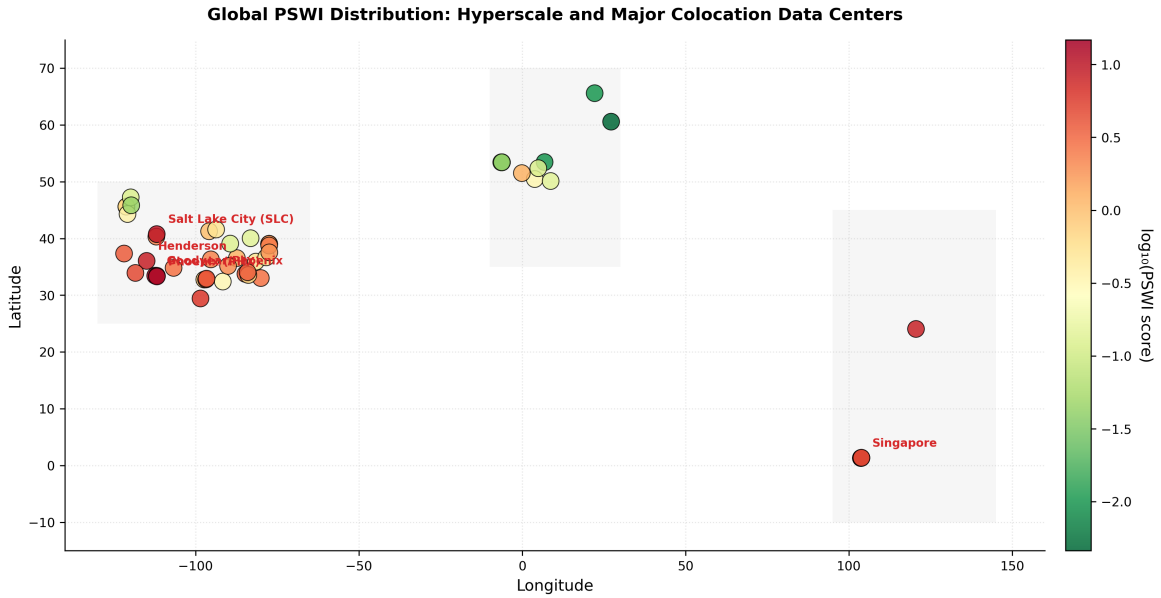

Fig. 7. Global distribution of audited facilities, colored by $\log_{10}(\text{PSWI})$. The U.S. Southwest and Singapore/Taiwan emerge as high-PSWI clusters; Northern Europe and the U.S. Pacific Northwest are uniformly low.

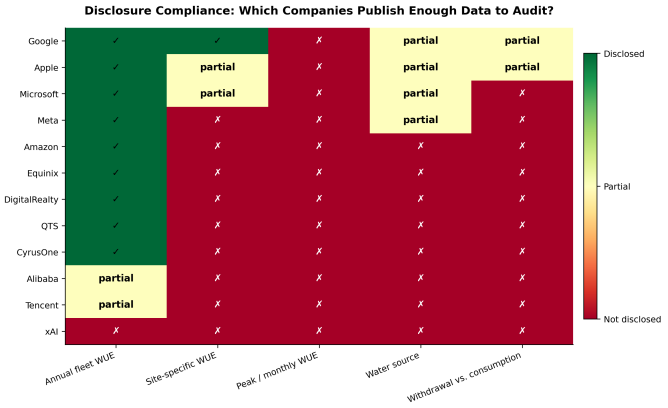

Fig. 8. Disclosure compliance heatmap for major data center operators. Only Google publishes site-specific WUE; *no* operator publishes peak or monthly water data, the metric most relevant for public-water- system stress assessment.

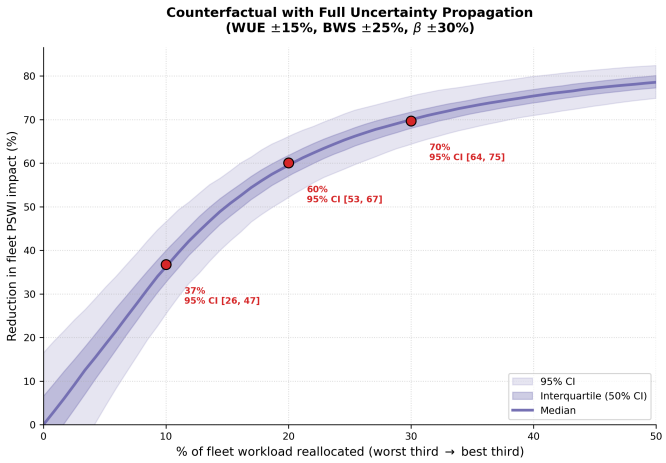

Fig. 9. Counterfactual impact reduction under PSWI-aware workload reallocation, with documented uncertainty propagated through all three inputs (WUE $\pm 15\%$, BWS $\pm 25\%$, β $\pm 30\%$). At 10% fleet reallocation: median reduction 36.7%, 95% CI $[26.1\%, 47.0\%]$.

concentration of demand and spatial heterogeneity of stress. We have introduced PSWI to integrate both, applied it to the first cross-company global dataset of major data centers, and documented robustness-tested findings: a 4,100$\times$ disparity between best and worst facilities; a same-operator 17$\times$ ratio that controls for reporting methodology; rank stability under $\pm 50\%$ β perturbation; and a counterfactual showing that 10% workload reallocation yields a 36.7% (95% CI $[26.1, 47.0]$) reduction in fleet impact under documented uncertainty in all inputs. We have proposed a regulatory framework engaging directly with the SEC's actual climate disclosure architecture. All data and code are publicly released.

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